Trial SET #3 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

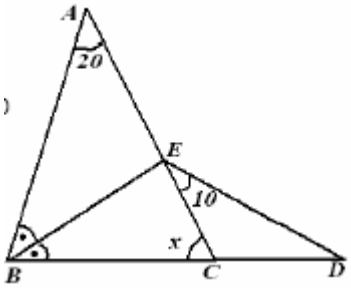
- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

Name of the student _____

Provided by 'skystudents.kz'

1. Kelly used to get a 30% discount on movie tickets. When the price of the movie ticket increased by 50%, the amount of discount in dollars remained the same. What is Kelly's discount with the new Ticket price in percent terms?
(a) 10%
(b) 20%
(c) 25%
(d) 35%
(e) 38%

2. In the given triangle ABC $\angle A = 20^0$, $\angle CED = 10^0$ and BE is angle bisector. If ED = BE, find $\angle ACB$?



A. 55; B. 60; C. 65; D. 70; E. 75;

3. $\frac{3^x + 3^x + 3^x}{6^x + 6^x + 6^x} = ?$

A) 2^x B) 2^{-x} C) 3^x D) 3^{-x} E) $\frac{1}{2}$

4. There are seven vocalists, four guitarists, four drummers, and two bassists in Glen Oak's music program, while there are five vocalists, eight guitarists, two drummers, and three bassists in Belmont's music program. If a band comprises one vocalist, one guitarist, one drummer, and one bassist, how many more bands can be formed in Belmont?

A. 4; B. 10; C. 16; D. 18; E. 26;

5. If $0 < x < 90$ and $\frac{\sin x + \cos x}{\sin x - \cos x} = 3$. Find tan x.

A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) 1 E) 2

6. $\sqrt{2} \cdot \sqrt[3]{4} \cdot \sqrt[6]{32} = ?$

A. 2; B. 4; C. $\sqrt[4]{4}$; D. $\sqrt[11]{8}$; E. $\sqrt[5]{16}$;

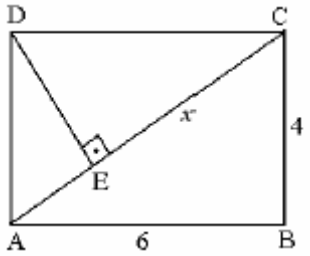
7. Following an increase in prices, the price of a candy box was 10 pounds and the price of a can of soda was 6 pounds. If the price of a candy box was raised by 25%, and the price of a can of soda was raised by 50%. What was the price of a box of candy plus a can of soda before prices were raised?

A. 11; B. 12; C. 13; D. 14; E. 14.5;

8. Consider the sequence 1, 2, 5, 1, 2, 5, 1, 2, 5, … What is the sum of the first 100 terms?

A. 220 B. 280; C. 298; D. 265; E. 264;

9. ABCD is a rectangle, $AC \perp DE$, AB = 6 cm. If BC = 4 cm then find EC .

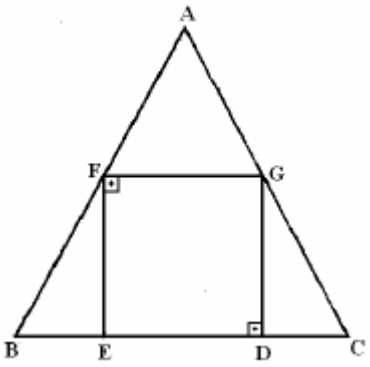


A. $\frac{9}{\sqrt{13}}$; B. $\frac{10}{\sqrt{13}}$; C. $\frac{12}{\sqrt{13}}$; D. $\frac{16}{\sqrt{13}}$; E. $\frac{18}{\sqrt{13}}$;

10. log (a + b) = log a + log b ➔ b = ?

A) $\frac{a}{a+1}$ B) $\frac{a+1}{a}$ C) $\frac{a}{a-1}$ D) a - 1 E) $\frac{a+1}{a-1}$

11. ABC is an equilateral triangle and DEFG is a square and area of triangle AFG = $5\sqrt{3}$ cm². Then find area of DEFG .



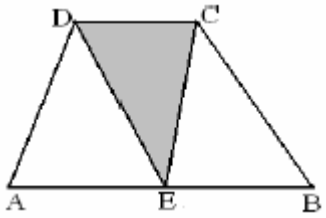
A. 28; B. $20\sqrt{3}$; C. 20; D. $\frac{25\sqrt{3}}{4}$; E. 30;



12. The figure above shows the remaining pieces of a paper chain from which a section has been removed. The first link of the original chain was red and the colors of the links formed the repeating pattern of red (R), yellow (Y), green (G), white (W), and blue (B) from left to right. Which of the following could be the number of links in the section that was removed?

(A) 60
(B) 61
(C) 62
(D) 63
(E) 64

13. ABCD is a trapezoid and DC = c, AB = a, a = 5c, then find the ratio of the area of ABCD to the area of DCE.



A. 3; B. 4; C. 5; D. 6; E. 7;

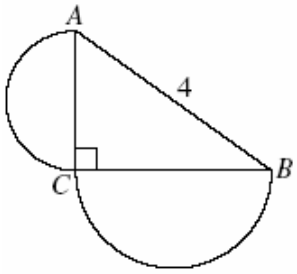
14. What is the smallest and greatest value of $|\cos\frac{\alpha}{2}| + 1$?

A. 0;2 B. 1;2 C. –1;2 D. –1;1 E. 0;1

15. Muammar must divide a certain number by 4 and add 10 to the result. He mistakes and divide the number by 2 and add 5. What should he do with his wrong result to obtain right answer?

A. divide by 2 and add 5; B. add 5 and divide by 2; C. multiply by 2 and add 2.5;
D. divide by 2 and add 7.5; E. multiply by 2 and subtract 2.5;

16. In the figure above, the legs of right triangle *ACB* are diameters of the two semicircles. If AB = 4 what is the sum of the areas of the semicircles?



A. π; B. 2π; C. 4π; D. 8π; E. 16π;

17. The original price of a book was b dollars. This price was discounted x percent during a sale. James used a coupon to buy the book for t percent off the sale price. Which of the following represents the price, in dollars, that James paid for the book?

(A) $b\left(1-\frac{x}{100}\right)\left(1-\frac{t}{100}\right)$

(B) $b\left(\frac{(1-x)(1-t)}{100}\right)$

(C) $b\left(1-\frac{x+t}{100}\right)$

(D) $b\left(1-\frac{xt}{100}\right)$

(E) $b\left(\frac{x}{100}\right)\left(\frac{t}{100}\right)$

18. The average (arithmetic mean) of 4 positive integers is 50. If the average of 2 of these integers is 45, what is the greatest possible value that one of the other 2 integers can have?

A. 55; B. 65; C. 100; D. 109; E. 115;

19. If $\log_3 a + \log_9 b = 2$ find the smallest value of $a + b$.

A) 4 B) 8 C) 9 D) 10 E) 12

20. A researcher plans to identify each participant in a certain medical experiment with a code consisting of either a single letter or a pair of distinct letters written in alphabetical order. What is the least number of letters that can be used if there are 26 participants, and each participant is to receive a different code?

A. 4
B. 5
C. 6
D. 7
E. 8

21. Which of the following definitions best fits to the graph of $y = e^{-x}$, where x takes values from -10 to 10.

A. falling tower; B. zigzag; C. mountains; D. plane flying up; E. landing plane ;

22. Find dy/dx if y = $\sin(\sqrt{2x})$;

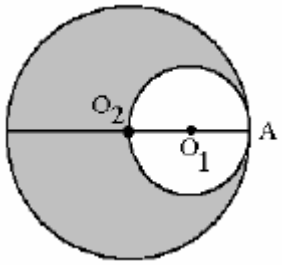
A. $\frac{\cos(\sqrt{2x})}{\sqrt{2x}}$; B. $\frac{\cos(\sqrt{2x})}{2\sqrt{2x}}$; C. $\frac{\sin(\sqrt{2x})}{\sqrt{2x}}$; D. $\sqrt{2x}\cos(\sqrt{2x})$;
E. $\frac{\cos(\sqrt{2x})}{\sqrt{x}}$;

23. If c is a constant, what is the value of $\int (2x+3)^3 dx$?

A. $\frac{(2x+3)^4}{4}+C$; B. $\frac{(2x+3)^4}{8}+C$; C. $4(2x+3)^4+C$;
D. $\frac{(2x+3)^4}{2}+C$; E. $\frac{(2x+3)^4}{16}+C$;

24. Observe that $17 = 4^2 + 1^2 + 0^2$. How many of the first 14 positive integers can be written as the sum of the squares of three integers ?
A. 10; B. 9; C. 14; D. 13; E. 12;

25. The circles centered at O_1 and O_2 are internally tangent to the each other at point A, find the ratio of shaded area to the area of circle centered at O_1.



A. 1; B. 3/2; C. 2; D. 5/2; E. 3;

26. Find the derivative of the function y = $\frac{\sqrt{2x+4}}{2}$

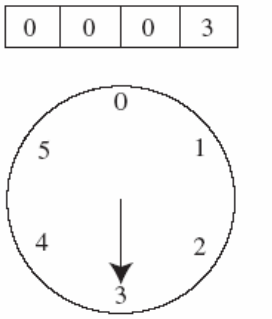
A. 1/y; B. 1/(2y); C. -1/(4y);
D. 2/y; E. 1/(4y);

27. *a*, *b*, and *c* are integers and $a < b < c$. *S* is the set of all integers from *a* to *b*, inclusive. *Q* is the set of all integers from *b* to *c*, inclusive. The median of set *S* is (3/4)*b*. The median of set *Q* is (7/8)*c*. If *R* is the set of all integers from *a* to *c*, inclusive, what fraction of *c* is the median of set *R*?
(A) 3/8
(B) 1/2
(C) 11/16
(D) 5/7
(E) 3/4

28. If $y = \frac{x^2 - 3x - 10}{x+2}$ then find dy/dx?

A. $\frac{3x+4}{(x+2)^2}$; B. 1; C. $\frac{x^3+4x+4}{(x+2)^2}$; D. $\frac{x^2+4x-4}{(x+2)^2}$; E. x-5;

29. The figure below shows a digital counter above and a dial counter showing the digits 0 through 5. Both counters are initially set to 0 and count upward together in increments of 1. For instance, when the digital counter reads 5 the dial counter also reads 5, but when the digital counter reads 6, the dial counter resets back to 0. What will the dial counter read when the digital counter reads 2000?



A. 0; B. 1; C. 2; D. 4; E. 5;

30. Solve the inequality: $2\cos^4 x - 3\cos^2 x + 1 > 0$.

A) $\pi k + \frac{\pi}{4} < x < \frac{3\pi}{4} + k\pi$ B) $\frac{\pi}{3} + 3k\pi < x < \frac{3\pi}{4} + k\pi$

C) $\frac{2\pi}{3} + k\pi < x < \frac{3\pi}{4} + k\pi$ D) $\frac{\pi}{6} + 2\pi k < x < \frac{5\pi}{6} + 2\pi k$

E) $\pi k \le x \le \frac{\pi}{4} + \pi k$